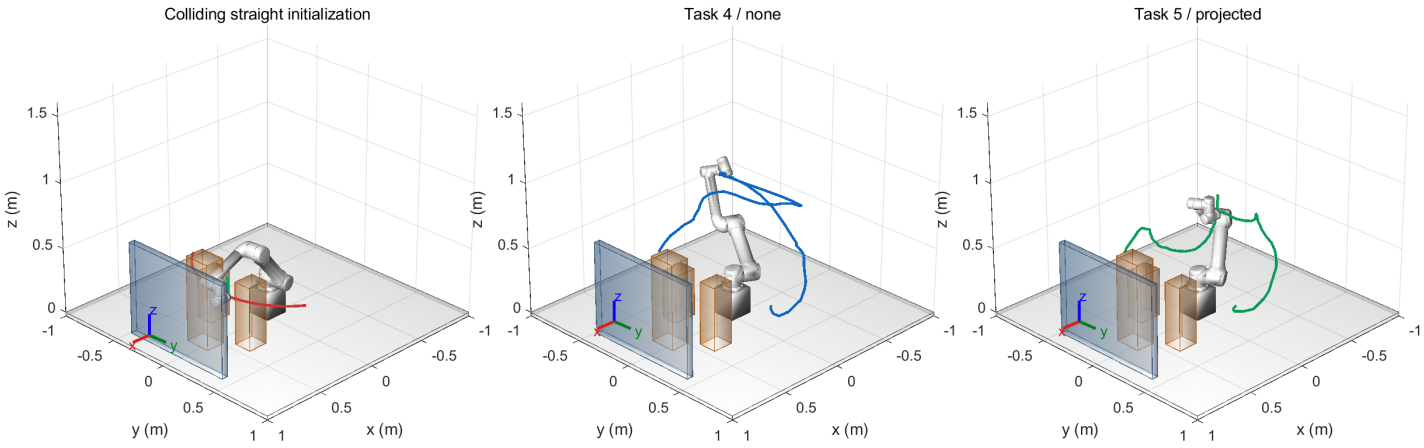
Task 4-5: mounted-workcell evaluation

V3 addendum / EE5112 Project 2 / 9 November 2026

A third scene tests the existing V2 planner with a physical mounting arrangement, an explicit table and a new endpoint pair. The fixed pedestal raises the UR5 base 0.20 m above the table. A central rack, two equipment boxes and a rear guard obstruct the transfer.



The straight joint-space initialization collides at 92/181 checked states. Task 4 and Task 5 share this initialization. The geometric feasibility witness is separate from every optimizer initialization and restart.

Mounting and collision scope

The table top is world z=0. The only permitted world contact is the fixed pedestal/table pair. No UR5 link or tool is exempt. The pedestal participates in actual self-collision checks and the optimization self proxy; every original UR5 sphere still requires 35 mm conservative world clearance.

New endpoints (rad): start [-1.10, -1.05, 1.35, -0.30, -pi/2, 0]; goal [1.05, -0.75, 1.10, -0.35, -pi/2, 0]. Both endpoints are upright and pass collision checks.

The saved scene box array drives the voxel field, actual collision objects and rendering. The sub-table forbidden volume covers the full voxel workspace below z=0.

Reserved-seed formal results

Development seeds 5300-5302 pass 6/6 and select the first profile. Formal seeds 5400-5419 are reserved until freezing: 20 seeds x 2 conditions = 40 runs. All runs enter the denominator. There are no failed formal runs in this study.

Condition	Success	Median time (s)	Worst tilt (deg)	Min gap (mm)	Min tool z (mm)
Task 4	20/20	30.42	169.591	35.41	273.1
Task 5	20/20	68.19	0.997	35.23	280.9

Tilt, gap and height are extrema over successful formal paths in the additional 0.00625 rad verification. Planner time is the median of all 20 runs per condition and includes trace writes. Conditions share the same two-worker execution setup.

Constraint and acceptance

Both groups use the V2 self-collision penalty and in-loop actual mesh rejection. Task 5 adds J_ori = 1000 sum ||R(q)e_y - e_z||_1 Delta s and the PoE direction projector. The tool y axis follows world +z while yaw remains free. Feasible candidates take priority over infeasible lower-cost candidates.

Both groups preserve exact endpoints and joint limits, require 35 mm proxy clearance, zero unpermitted world/self collisions and tool above the table. Task 5 also requires maximum tilt <=5 degrees. Final validation uses 0.0125 rad joint increments; every successful formal path also passes 0.00625 rad.

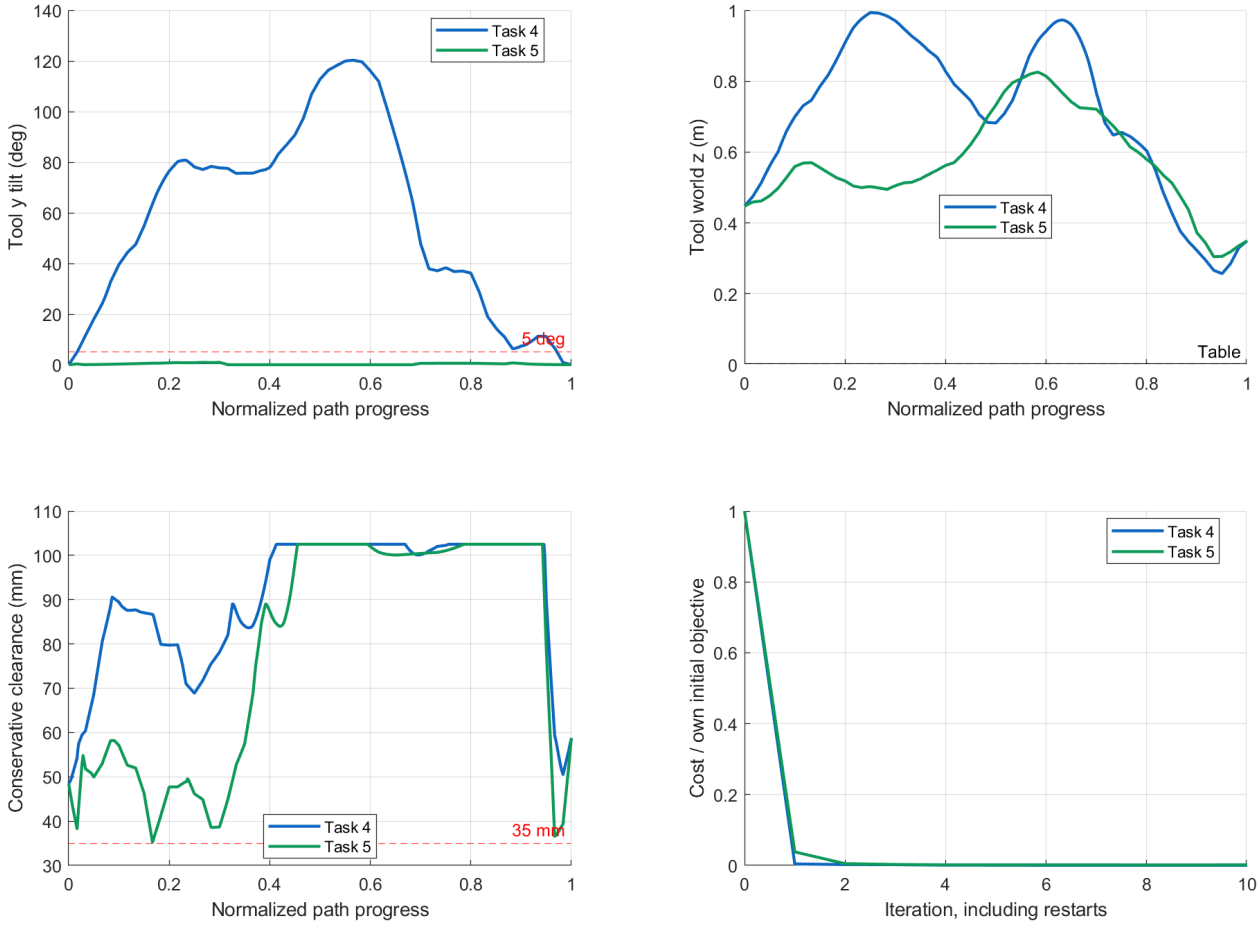
Common successful pairs

n=20; median paired planner time (Task 5 minus Task 4) 37.19 s. Different weighted objective totals are not used as a quality comparison.

For 20/20 successes, the exact two-sided 95% binomial lower bound is 83.16% assuming independent Bernoulli trials in this fixed scene. This does not establish general reliability, and paired conditions are not pooled.

Member C adds the workcell, mount/contact policy, table-height gate, formal paired study and evidence. The team PoE, sphere envelopes and STOMP stages are reused. V1/V2 records remain intact.

Evidence, scope and reproduction



Demonstration seed 5300. Blue: Task 4; green: Task 5. The table and orientation gates are shown against normalized path progress. Each objective curve is divided by its own initial value. Progress is not execution time.

The independent audit recreates the mounted model and checks all 40 formal paths, including 804 optimizer checkpoint replays. The two demonstrations add 20 replayed checkpoints. A separate NumPy FK cross-check also matches all 46,798 fine states (position difference <2e-15 m); actual collision checks remain in MATLAB.

V2 had no floor: 7/20 projected formal paths placed the tool below base z=0. V3 addresses that modeling limitation. Changed endpoints and seeds prevent causal cross-version performance claims. Checks remain discrete; payload geometry, dynamics and real robot execution are outside this study.

MATLAB replay: addpath('scripts'); show_project_demo('all','v3'). Fresh planning: run_task45_v3(...); independent verification: verify_task45_v3(runRoot). See README_TASK45_V3.md and docs/TASK45_V3_RESULTS.md for configuration and archive scope.